FDE SCENARIO DRILLS #05

FDE SCENARIO DRILLS · ISSUE 05 OF 08

The CDC slot that filled the disk.

Scenario: the source database ran out of storage on a Sunday.

05

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Swipe →

How it happened

1. CDC consumer crashes
2. Replication slot stops advancing
3. PostgreSQL keeps the WAL
4. WAL grows for days
5. Disk full, writes fail

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Recover

01 **Free space safely**

Restart the consumer, or drop the slot with the DBA's approval.

02 **Resnapshot**

If the slot is dropped, take a new consistent snapshot.

03 **Reconcile**

Row counts and checksums between source and target.

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Make CDC safe

01 **Alert on slot lag**

Retained WAL size, well before the disk limit.

02 **Cap retained WAL**

max_slot_wal_keep_size where the service allows it.

03 **Idempotent apply**

Upsert by key, skip changes with an older LSN.

04 **Own the runbook**

Agree with the DBA who acts, and when.

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THE LESSON

Your pipeline can take down their database.

A stalled reader is a production risk on the source system.

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What interviewers listen for

- [x] Understands WAL and slots
- [x] Involves the customer DBA
- [x] Plans for duplicates and gaps
- [x] Monitors the source, not just the target
- [x] Reconciles after recovery

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KEY TAKEAWAY

Monitor the source database, not just your pipeline.

→ Alert on replication slot lag

→ Make the apply step idempotent

→ Agree the recovery runbook upfront

YOUR TURN

Have you ever had a pipeline harm the system it reads from?

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